UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

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REG NO	*23MDBCS435*
SECTION	*A*
DEPPT	*COMPUTER SCIENCE*
SUBJECT	*DLD LAB*
TEACHER	*SHEHZAD SIR*

LAB TASK : 3

XOR Gate

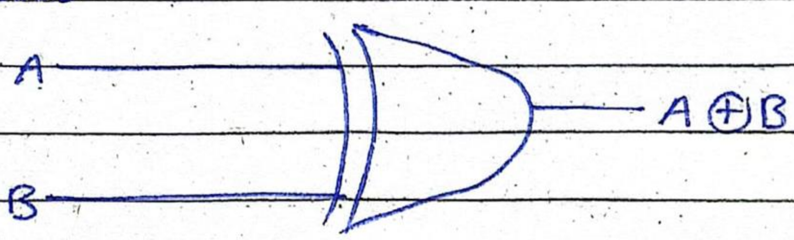
XOR Gate pronounced as exchange OR-gate is a digital logic gate that give 1 output when the number of true inputs is odd.

Truth Table

Input

Input	Input	Output
0	0	0
0	1	1
1	0	1
1	1	0

Diagram



Applications

1. Arithemitic Operations

The XOR Gate is also called the Modulo two adder, since it give the sum of two binary numbers.

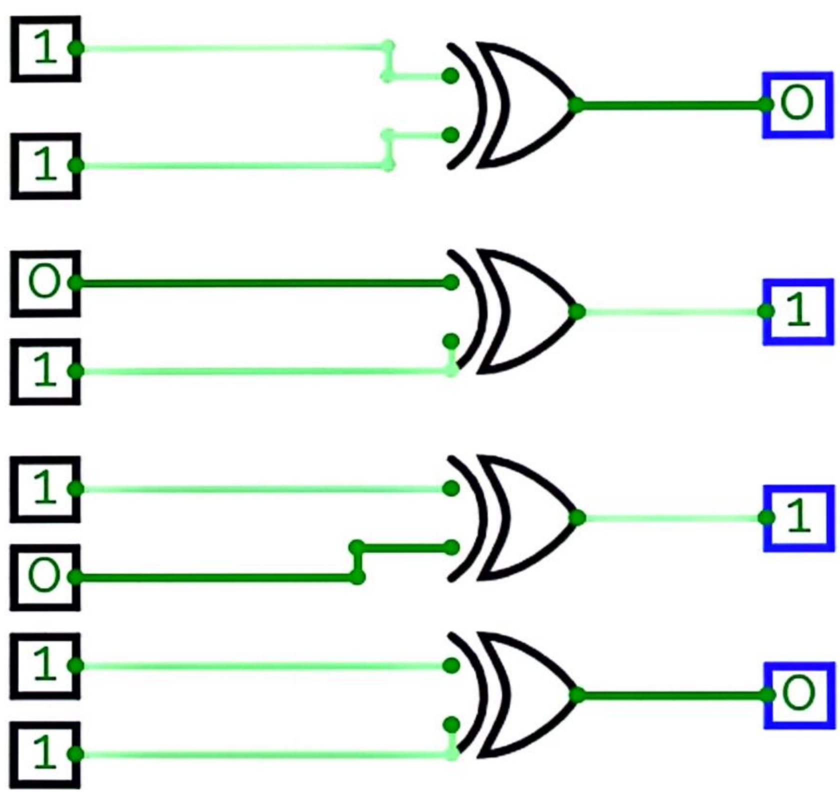
2. Parity Checker

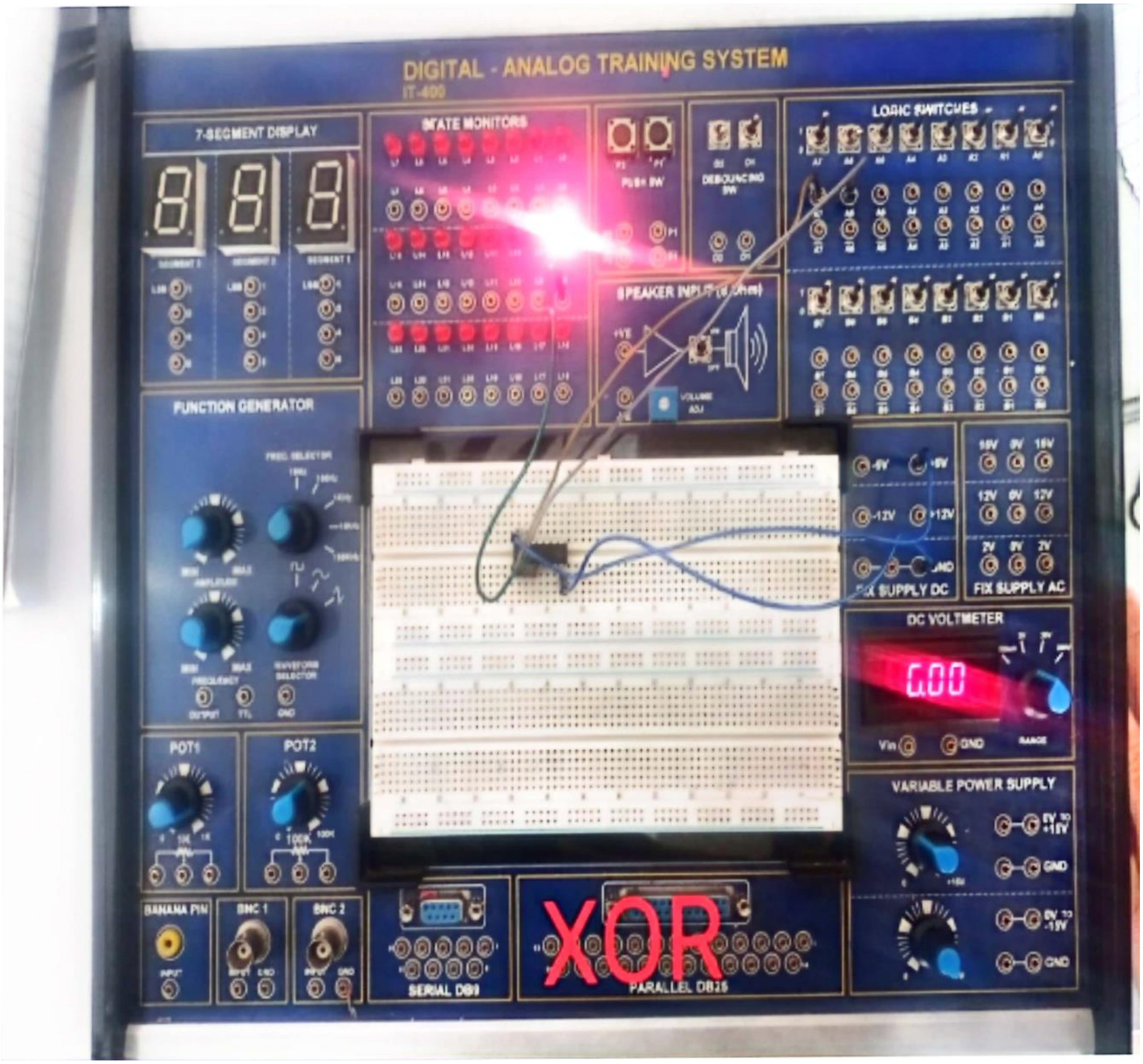
One one of the simplist form of error detection is parity checker. The XOR gate is the most suitable circuit to provide parity checker.

3. Control Inverter

The XOR Gate can be used as Not Gate by connecting one of the inputs to logic (1).

→ There are many others uses of XOR Gate as well.




DIGITAL - ANALOG TRAINING SYSTEM
7-SEGMENT DISPLAY
STATE MONITORS
LOGIC SWITCHES
FUNCTION GENERATOR
SPEAKER INPUT
FIX SUPPLY DC
FIX SUPPLY AC
DC VOLTMETER
0.00
POT1
POT2
VARIABLE POWER SUPPLY
BANANA PIN
BNC 1
BNC 2
SERIAL DB9
XOR
PARALLEL DB25

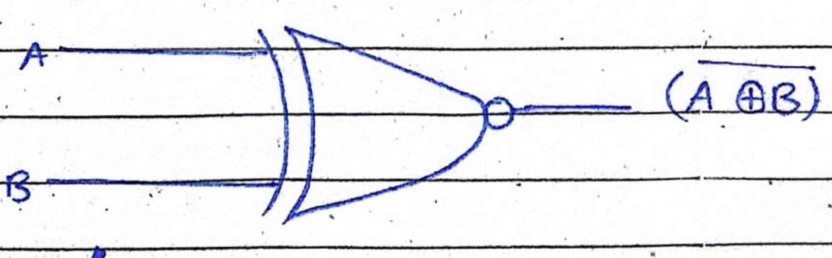
X. NOR - Gate

The X-NOR Gate is the logical complement of Exclusive OR gate.
It produces high (1) output on the similer Inputs.

Truth Table

Input		Output
0	0	1
0	1	0
1	0	0
1	1	1

Symbol



Applications

a) Computer Architecture:

X-NOR Gates are used in computer Architecture to

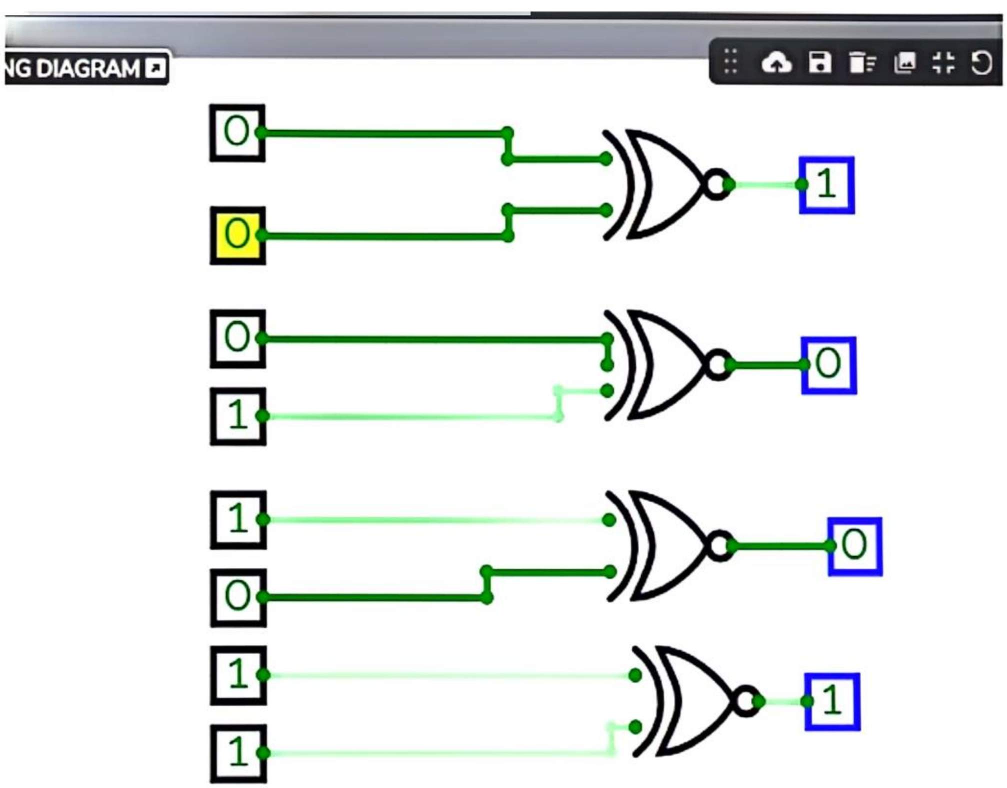
To Perform Arithemitic and logical operators.

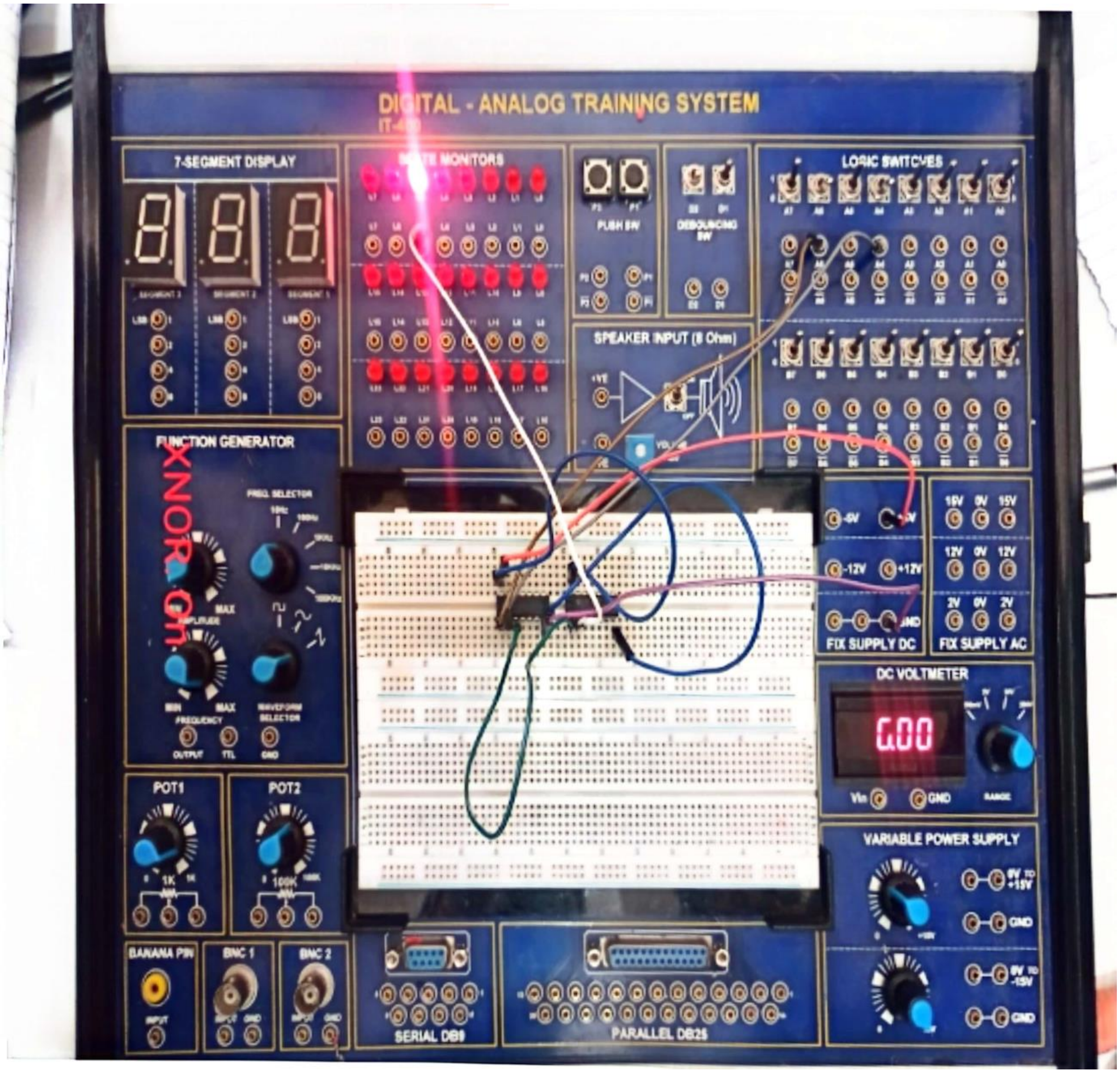
b) Computer Security

X-NOR Gate finds applications in computer security systems such as access control mechanism.

c) Computer Networks

X-NOR gates are used in computer networks for various purposes including routing algorithems and network protocals.




DIGITAL - ANALOG TRAINING SYSTEM
7-SEGMENT DISPLAY
FUNCTION GENERATOR
XNOR on
LOGIC SWITCHES
SPEAKER INPUT (8 Ohm)
DC VOLTMETER
VARIABLE POWER SUPPLY
FIX SUPPLY DC
FIX SUPPLY AC
POT1
POT2
BANANA PIN
BNC 1
BNC 2
SERIAL DB9
PARALLEL DB25

THE END